INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

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THE SITE

Location	TX facility TX_way_1498670679 -- station KTPL, America/Chicago
Building	OSM 1498670679 OSM way 1498670679
Facade gap	not applicable -- one building, no second facade
Weather record	43,374 real hourly records from KTPL
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-11-03 (clear_cool)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 8 of 24 hours with 1 mode change. That is 4 more than the reactive incumbent, at 0 unsafe hours against its 0. 4 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	8 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	4 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 4 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

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hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dew point	12.780	18.0	13.890	4.464
01	mechanical	no	dew point	12.780	18.0	13.890	3.480
02	mechanical	no	dew point	13.330	18.0	13.330	3.219
03	mechanical	no	dew point	13.340	18.0	13.330	2.820
04	mechanical	yes	switch budget	13.890	18.0	12.220	3.036
05	mechanical	yes	switch budget	13.890	18.0	11.670	2.689
06	mechanical	yes	switch budget	14.440	18.0	11.110	2.888
07	mechanical	yes	switch budget	15.000	18.0	11.110	2.351
08	mechanical	no	dew point	17.220	18.0	10.000	2.586
09	mechanical	no	dry-bulb	21.110	18.0	10.000	3.909
10	mechanical	no	dry-bulb	23.880	18.0	9.440	5.563
11	mechanical	no	dry-bulb	25.560	18.0	9.440	6.057
12	mechanical	no	dry-bulb	26.670	18.0	9.440	6.407
13	mechanical	no	dry-bulb	26.670	18.0	9.440	6.216
14	mechanical	no	dry-bulb	22.230	18.0	9.440	4.480
15	mechanical	no	dry-bulb	18.890	18.0	9.440	3.161

16	FREE	yes	-	15.560	18.0	9.440	2.544
17	FREE	yes	-	13.330	18.0	9.440	2.693
18	FREE	yes	-	11.110	18.0	8.890	2.904
19	FREE	yes	-	8.890	18.0	8.890	2.949
20	FREE	yes	-	7.770	18.0	8.890	3.602
21	FREE	yes	-	7.220	18.0	8.890	3.993
22	FREE	yes	-	7.220	18.0	8.330	4.381
23	FREE	yes	-	7.770	18.0	8.330	4.646

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 12.780 C would have passed. The outside air is too HUMID: the dew-point bound is 15.55 C against a 15.0 C maximum, failing by 0.55 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

01:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 12.780 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

02:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 13.330 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

03:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 13.340 C would have passed. The outside air is too HUMID: the dew-point bound is 15.01 C against a 15.0 C maximum, failing by 0.01 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.890 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.890 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.440 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.000 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 17.220 C would have passed. The outside air is too HUMID: the dew-point bound is 15.01 C against a 15.0 C maximum, failing by 0.01 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.110 C against a 18.0 C limit, so it fails by 3.110 C. It would take a limit 3.110 C higher, or a bound 3.110 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.880 C against a 18.0 C limit, so it fails by 5.880 C. It would take a limit 5.880 C higher, or a bound 5.880 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.560 C against a 18.0 C limit, so it fails by 7.560 C. It would take a limit 7.560 C higher, or a bound 7.560 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.670 C against a 18.0 C limit, so it fails by 8.670 C. It would take a limit 8.670 C higher, or a bound 8.670 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.670 C against a 18.0 C limit, so it fails by 8.670 C. It would take a limit 8.670 C higher, or a bound 8.670 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.230 C against a 18.0 C limit, so it fails by 4.230 C. It would take a limit 4.230 C higher, or a bound 4.230 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.890 C against a 18.0 C limit, so it fails by 0.890 C. It would take a limit 0.890 C higher, or a bound 0.890 C tighter, to change this hour.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.560 C, which is 2.440 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.544 C, and the margin is measured, not chosen: 2.544 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.330 C, which is 4.670 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.693 C, and the margin is measured, not chosen: 2.693 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.110 C, which is 6.890 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.904 C, and the margin is measured, not chosen: 2.904 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.890 C, which is 9.110 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.949 C, and the margin is measured, not chosen: 2.949 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.770 C, which is 10.230 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.602 C, and the margin is measured, not chosen: 3.602 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.220 C, which is 10.780 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.993 C, and the margin is measured, not chosen: 3.993 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.220 C, which is 10.780 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.381 C, and the margin is measured, not chosen: 4.381 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.770 C, which is 10.230 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.646 C, and the margin is measured, not chosen: 4.646 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

WHAT IT IS WORTH OVER FIVE REAL YEARS, AT THIS SITE

Held-out days simulated	912 -- the agent never calibrates on a day it is scored on
Free cooling delivered	8.46 h/day, i.e. 3,088 h/yr
Chiller-hours avoided	+340.4 h/yr against a tuned reactive incumbent
12-hour plan stability	94.8 % of 21,565 re-plans change nothing at all

The ladder, one constraint at a time:

N-56-like: notice 0, skill 1.00, no constraint	+81.3 h/yr
+ switch budget 2, min dwell 3 h	+106.1 h/yr
+ dew-point gate 15 C (Green Grid WP#46 p.6)	+101.3 h/yr
+ notice 3 h, skill 0.50 (no perfect forecast)	+340.4 h/yr
+ unanchored, 4 measured FG offsets rotated	-155.4 h/yr

PRICED, IN THIS STATE'S OWN ELECTRICITY TARIFF

Chiller power	156.1 kW per MW of IT load (0.549 kW/ton, the ASHRAE 90.1-2019 minimum)
Electricity	8.72 cents/kWh (Virginia commercial, 2024 annual)
Energy avoided	53,141 kWh per MW of IT load per year
Value	$4,634 per MW of IT load per year

Everything above is PER MEGAWATT OF IT LOAD. This project has never measured a data centre's size and will not invent one; a reader who knows their own IT load multiplies once.

WHAT IS NOT CLAIMED

- The chiller COMPRESSOR only. Fans, chilled-water pumps, condenser pumps and cooling-tower fans keep running, and an airside economizer moves MORE air, so fan power can rise. The unmeasured term has the OPPOSITE SIGN, so this is an upper bound.
- Code-minimum chiller efficiency is the OPTIMISTIC end: Standard 90.1 is a floor, hyperscale plants beat it, and a better chiller saves less per hour switched off.
- Full-load kW/ton OVERSTATES the draw at the moment free cooling is available, because free cooling happens when it is cool and a chiller at a cool condenser runs below its rating. IPLV is the lower published number but is an annual-weighted metric under AHRI's assumed load profile, not the part-load point free-cooling weather sits at.
- The heat rejected is APPROXIMATED by the IT load. UPS, PDU and lighting losses inside the envelope add to it; some overhead sits outside the cooled space, so scaling by full PUE would overshoot. No PUE multiplier is applied.
- EIA STATE-SECTOR AVERAGES, not the site's tariff. A Loudoun County campus buys on a large-general-service contract that is not public.
- No carbon, no water, no maintenance and no demand charges. Tower water use moves the OTHER way when the chiller runs less.
- Every caveat on the hours is inherited: the headline is conditional on the bank sitting on the long facade (the refusal guard is priced at -3,124 h/yr where it fires) and the unanchored row is NEGATIVE. Those rows appear here with their signs intact.

HOW TO REPRODUCE EVERY NUMBER IN THIS REPORT

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cd INTAKE-ARBITER/src && python run_all.py
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That rebuilds every artefact from saved data and then audits it: 39 checks, 68 published figures re-read out of the files the code itself wrote, and it exits non-zero on any failure. It makes ZERO API calls -- every input is a saved FortyGuard response, a committed geometry file, or the station's own hourly record.

Generated by INTAKE-ARBITER/src/report.py. Verified by being read back: the file was reopened after writing and every hour of the schedule, the headline counts and this site's own name were confirmed present.